Gabriel Thien

Embedded Software / Mechatronics Engineer @ Cochlear Ltd.

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Australia

LinkedIn

GitHub Repositories

SUMMARY

Curious problem-solver with a enthusiastic, non-blocking mindset for building things. Loves spreadsheets, efficiency and making things easier and flexible. Engineer with diverse experience in embedded software, automation & mechatronic systems, MCUs, IoT and product development. Team-player who loves tinkering in the workshop, owning and developing end-to-end solutions with low-level electronics, electromechanical systems and firmware. Looking to relocate to Melbourne, Victoria.

EDUCATION

UNSW Sydney - (2021 ~ 2025)

B. Engineering (Mechatronics) & B. Science (Computer Science)

EXPERIENCE

Cochlear Ltd - (Nov 2024 - Present)

Cochlear is the global leader in implantable hearing devices used by over 700,000 people worldwide. Implants are built for lifetime reliability and Cochlear's engineering standards for firmware architecture, low-power electronics and fine manufacturing are extremely rigorous to meet medical-device regulations, making their products sit above typical consumer electronics.

Graduate Embedded Software / Mechatronics Engineer

- Rotating across manufacturing and R&D, building deep and varied industry knowledge
- Architecting, writing and debugging firmware with custom RTOS and kernel control, tooling in Python, C#, DevOps
- Orchestrating automated electrical test equipment with National Instruments, developing cobot automation
- Debugging embedded electronics with oscilloscopes and logic analysers, handwork
- Undergoing technology exploration and infrastructure improvements to workplace

Associate Firmware Engineer

- Enhanced RTOS firmware development by developing stack analysis tools (Python/C), allowing for granular memory usage view of programs and tasks on top of Chess compiler toolchains
- Reduced complexity of in-house ASIC & firmware test systems with gRPC architecture
- Secured highly competitive graduate engineer position for 2026, rotating through various departments

Asiga 3D Printers - (Mar 2023 - Nov 2024)

Asiga is a leading manufacturer of high-resolution resin 3D printers, serving dental and industrial applications, founded on the world's first LED-based DLP 3D printer launched in 2011. Its printers rely on embedded systems for precise LED/UV exposure control, layer thickness control, closed-loop layer positioning and auto-calibrating optical hardware, all tuned for micron-level repeatability and precision across a range of resin materials.

Associate Mechatronics Engineer

- Developed bespoke electro-mechanical systems used in manufacturing and quality control with embedded Linux, ESP32
- Designed debugged jigs & fixtures, electronics circuitry and STM32 firmware, desktop control software with Qt
- Developed embedded electronics and control software for motors and sensors with C/C++, ESP32 MCUs, using communication protocols such as I2C, UART, SPI

PROJECTS

Logistic & Resource Scheduling System - Factorio (2026)

- Designing a many-to-many, dynamic optimal resource train scheduling system
- Simulation written in C to mimic Factorio's game mechanics

Common Desktop Controller App for MCUs

- Programming a desktop app that views and controls peripherals attached to MCU
- Displays stream data from MCU, loads new apps and writes memory blocks

INTERESTS

Bouldering. Hiking. Baking. Factorio. Mahjong.

TECHNICAL SKILLS

Embedded Systems

FreeRTOS , eChronos RTOS, Atmega, STM32CubeMX frameworks

PCB electronics design, TH & SMD soldering, characterisation, scope and logic analyser debugging

Mechatronics

System process design, fixtures, 3D printing, laser cutting, general workshop skills

Software Development

C/C#/C++, Python, Qt, .NET frameworks, Agile Development, Git, Linux